SONIFICATION AS CATALYST IN TRAINING MANUAL WHEELCHAIR OPERATION FOR SPORTS AND EVERYDAY LIFE

Andreas Almqvist Gref
KTH Royal Institute of Technology
aalmqvi@kth.se

Ludvig Elblaus
KTH Royal Institute of Technology
elblaus@kth.se

Kjetil Falkenberg Hansen
KTH Royal Institute of Technology
Södertörn University
kjetil@kth.se

ABSTRACT

In this paper, a study on sonification of manual wheelchair movements is presented. The aim was to contribute to both rehabilitation contexts and in wheelchair sports contexts, by providing meaningful auditory feedback for training of manual wheelchair operation. A mapping approach was used where key parameters of manual wheelchair maneuvering were directly mapped to different sound models. The system was evaluated with a qualitative approach in experiments. The results indicate that there is promise in utilizing sonification for training of manual wheelchair operation but that the approach of direct sonification, as opposed to sonification of the deviation from a predefined goal, was not fully successful. Participants reported that there was a clear connection between their wheelchair operation and the auditory feedback, which indicates the possibility of using the system in some, but not all, wheelchair training contexts.

1. INTRODUCTION

Our perception of sound is linked to our understanding of physical properties of objects, as well as our understanding of how objects interact and move in the physical world [1, 2]. In music, the gestural performance and its link to sound and bodily movement has been the subject for much research and has been shown effective by Wanderley among others [3,4], and performers' movements are moreover affected by the instant audio feedback from bodily interaction with the instrument in a closed-loop sonification [5].

Studies of sonification of body movements have shown that this type of feedback may improve motor task learning by making movement relations more obvious to the user of the system [6]. This can be done either by giving feedback on the deviation from a desired movement or by a more time demanding approach where the feedback is linked not to the deviation but to the movement in a more direct sense, and thus give guidance towards a goal which is apparent to the user by already present stimuli or by the assistance of a trainer.

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In the presented study the possibilities of auditory feedback for the purpose of training manual wheelchair movements are studied. The targeted training context is that of expert supervised exercises in a gym, where precision is valued higher than portability. There are two main fields that may be considered; firstly in everyday-life and rehabilitation contexts relating mostly to learning fundamental movement technique, and secondly, movement in wheelchair sports. Our aims are partly to investigate sonification as catalyst and motivation in repetitive training, and partly to investigate audio feedback in training of accurate movements where other bio-feedback is insufficient.

Manual wheelchairs are used by a variety of persons who are partly or fully dependent on an assistive device for locomotion. The only property these users share are that they are able to use their arms for propelling the wheelchair. While standard wheelchairs may be similar in features, the group of mobile wheelchair riders have a smaller base of common ground than what might be the first assumption. Depending on the disability, operation will be done differently. For example, if the disability inhibits from leaning forward, as with high positioned spinal cord injury, both wheelchair propulsion and the fundamental skill of back wheel balance becomes significantly more difficult [7].

The most basic control of a wheelchair is done through grabbing the push rings of the back wheels and pushing the wheels in synchrony to gain forward propulsion. Applying more force to one wheel than the other allows for turning. A wheelchair for everyday-life use needs to be able to cross obstacles such as door frames and curbs by tilting the chair backwards into a back wheel balancing position. Sport wheelchairs have different design constraints than that of the everyday-life wheelchair. Instead of front wheels, most of these sport chairs have small caster wheels, two in front and two in back. Back wheel balancing is still important however, with all four casters hovering slightly above ground. Maintaining this position eases both turning and propulsion. With good technique this will allow for turning without using hands on the push rings.

In this work the sonification definition proposed by Hermann [8] is used, which entails four major criteria: that the sound produced reflects *objective* properties of the input data, that the transformation to sound is *systematic*, and that this transformation is *reproducible* meaning that the output sound is consistent and structurally identical given the same input data. Additionally the sonification system must be *general* and possible to use for other data.

1.1 Strategies for Feedback

In relation to motor task learning, the sonification definition criteria of reproducible and systematic sonification is especially important. Without a consistent feedback the interaction loses its ability to aid in gaining insight [8–10]. Sonification of bodily movements or human activity in general may, if the above criteria is met, be used to clarify goal attainment without the explicit inclusion of goal specifics into the sound model [8]. The sound then acts as periphery guidance towards a self defined goal or a goal which, the attainment of, is evident by another modality. Hermann argues [11] that sonification of body movements may act as a way for the user of the system to gain insight into the movements performed. The sonification allows for monitoring of ones activities and consequently to evaluate differences in gesture execution.

Previous work show that sonification and auditory feedback improve motor task learning of complex movements [9,10] and that auditory feedback can enhance both perception accuracy and reproduction accuracy of complex sport movements [6]. Furthermore, the use of auditory feedback as a partial replacement for visual and sensory information has been shown effective for maintaining balance in upright stance [12], indicating that auditory bio feedback of deviations from a desired movement may help proprioception. Sigrist argues [10] that feedback for motor task learning should be designed as to direct the learner to the already present information that is most relevant for the execution of the movement. Sonification in therapy can also help to examine movements of a patient [13].

In summary, when learning complex movements or motor tasks, one must receive continuous feedback of the performance. The strategy of sonification becomes important because it must relate to the goal of the system. Sigrist outlines three main approaches [14]: 1) Auditory alarms, meaning discrete feedback when a predefined threshold is exceeded; 2) Sonification of movement variables, where movement continuously controls auditory feedback; and 3) Sonification of movement error, meaning continuous sonification of the deviation from a predefined movement pattern. Auditory alarms are discontinuous and less likely to stimulate complex motor task learning. Feedback on movement variables will depend on the user to have knowledge of a properly performed movement. Sonifying movement error will demand detailed knowledge of the characteristics of a correct movement to be built into the system.

No particular strategy from previous studies was adopted. Mapping continuous numerical values to pitch, time information to rhythmic patterning, key events to loudness and data concerning height to pitch height have some support by evaluation [14], but in general, few studies have systematically evaluated the efficiency of different sonifications [15].

1.2 Training in Manual Wheelchair Operation

According to Goosey-Tolfrey [16] the interaction between rider and wheelchair is complex and still not well understood, and overall performance depends on the ergonomics of wheelchair, the individual physical capacity, and driving technique. Furthermore, ergonomic adjustments and driving techniques vary between individuals, and is largely based on individual preference and self-learned adjustments. Modern physiotherapeutic methods for training include strategies for minimizing effort and maximizing control, derived from studies of mechanics of the wheelchair and bio-mechanics in the wheelchair-user configuration [7, 17], but they still remain somewhat imprecise. There is some consensus on methodological practices when teaching technique or evaluating capacity as to using generic exercises in testing [18, 19].

2. METHOD

2.1 Sonification System

The system consists of an *OptiTrack Prime 41* motion capture (MoCap), a computer running Pure Data, and eight (8) speakers set up in an octagon shape. [1] Spatialized sound is produced based on pre-defined movement parameters. An everyday-life wheelchair was equipped with three passive MoCap markers, one on each drive wheel axis and one on the right side of the wheelchair footrest. Four movement parameters were calculated from the MoCap, related to heading (direction), turning, speed, and tilting. [2] Table 1 shows the parameters and their mapping to the sonification models with link to sound examples.

Three different sonifications, or sound models, are named hereafter SM1–SM3. The SM1 is synthesized using additive synthesis of seven sine wave oscillators, with static proportion between oscillators. SM2 is synthesized using two mixed and continuously looped pre-recorded sounds. SM3 is synthesized using subtractive synthesis of white noise with four bandpass filters having center frequencies at odd harmonic intervals to produce a flute like sound. All sound models are spatialized to the heading using vector based amplitude panning, and they are used in two combinations: SM1+SM3 for sonifying turning and SM2+SM3 for sonifying speed.

2.2 Experiment

Four experiments sessions were conducted in this work, during which participants performed exercises derived from the methodologies of the book “Drivkraft” [7] and from works by Vereecken [19] and Inkpen [18]. The exercises were chosen to include basic movements at intermediate and advanced levels. After the exercise session, participants answered questions in a semi-structured interview.

For experiment session 1, one expert manual wheelchair user was recruited from a pre-study. The participant, male age=45, had actively used a manual wheelchair for 25 years and played wheelchair basketball at elite level. The participant reported having never heard a sonification before. For experiment sessions 2–4, novice subjects without disabilities were recruited (N=1, 1, and 4 respectively).

[1] Additionally, a plate with four pressure sensors was placed under the seat of the wheelchair. This was however not working optimally and has been omitted from the study as it did not affect the results.

[2] One parameter, related to leaning, was calculated from pressure sensors, but omitted as stated above.

Parameter	Description	Sonification	Mapping	Sound*
Heading	Direction of wheelchair	Spatialization	Panning	
Turning	Wheels rotational difference	SM1: Drone	Linear; fast turning –> high frequency	
Speed	Wheels rotational speed sum	SM2: Loop	Linear; high speed –> high frequency	
Tilting	Footrest vertical displacement	SM3: Flute	Linear; high position –> high frequency	

* sound examples: `https://soundcloud.com/user-61759848-282584838/sets/sound-examples-smc2016/s-wNAvs`

Table 1. Physical movement parameters extracted from motion capture and sensors, and their mapping to a sonification model. Links to sound examples are in the right column.

Exercise 1 drive in a figure-of-eight shape between cones separated by $1.5m$

Exercise 2 drive back and forth between cones separated by $5.5m$, alternate turns clockwise $180°$ and counter-clockwise $180°$

Exercise 3 drive in back wheel balance $3m$ across the room, passing two sets of cone-gates separated by $1m$

Exercise 4 propel forward and without further pushing the drive rings turn $120°$ around a cone and steer towards a goal cone

The four experiment sessions followed a similar procedure except for some changes: In session 1, the expert participant did an alternative version of exercise 3, driving backwards in back wheel balance; Session 1–3 were performed individually, while session 4 was a focus group study with additional discussion and demonstrations.

The participants were told what the technical components of the sonification system were and that turning and tilting [3] will produce auditory feedback. The participants were not told of the spatialization of sound nor of the variant with speed mapped to sound. The participants were then instructed too freely operate the wheelchair to gain familiarity with the system. Before proceeding to the exercises the participants were told to perform each exercise with speed and precision. It was emphasized that the goal of the exercises were primarily good precision, and secondary high speed. The participants were told to notify when satisfied with both precision and speed to proceed to the next exercise. There was no time limit imposed on the participants. After each exercise the participants were briefly reminded to execute the exercises with speed while focusing on precision.

Each exercise was explained to the participants before they performed it without any auditory feedback. After reporting being satisfied with the exercise execution the **turning sonification (SM1+SM3)** was activated and the participants continued to do the exercise until satisfied. Before moving to the next exercise the participants were allowed to question or comment. Exercises were performed successively in this manner. After completing all four exercises the sonification was switched to **speed sonification (SM2+SM3)** and the participants were told of the change and instructed to perform exercise 4 once more. Again, the participants were told to finish when satisfied with both speed and precision. After finishing all of the exercises the participants were asked a set of questions in a semi-structured interview.

[3] and in sessions 1 and 2 even leaning, but due to physiological and technical issues, this was omitted.



Figure 1. Picture shows a participant during execution of one of the exercises in the experiment.

3. RESULTS

Due to the selection of participants, the four experiment sessions were quite different even though the protocol was similar. In session 1, the participant was an expert rider, in sessions 2 and 3 the participants were inexperienced, while session 4 included group activities with an inexperienced focus group. In the following, the results will be presented accordingly.

3.1 Observations

The expert wheelchair rider, Male45, performed all exercises with ease and according to the instructions using less than 20 minutes to complete them. The participant did not express any difficulties with understanding the exercises. Exercises 1 and 2 were performed with the least variation, the participant did not ask any further questions after having been instructed. In exercise 3 the participant tried more variations as to how the exercise could be performed. The ultimate movement pattern, before proceeding to exercise 4, meant taking short and powerful strokes during back wheel balance with large variation in tilt. In exercise 4 the participant tried many variations on the turning. Since some room was given as to the approach to the cones, the instruction did not say to propel in a straight line, the par-

ticipant tried variations on the angle of the turn.

The novice participants executed the exercises in approximately 5 minutes per person and exercise. The participants in experiment sessions 2 and 3, Male28 and Male22, were able to complete all of the four exercises, although exercises 3 and 4 only after some practice. None of the participants in experiment session 4 could complete exercise 3 and two of the participants could not complete exercise 4 while the other two could.

3.2 Interview Summary Expert Participant

The participant reported hearing the auditory feedback clearly and that the sounds were relatively pleasant. When asked about which movements that generated sound, he said that he understood and added that the tipping sound (VDF-SM3) differed from the rest in being distinctly connected to the movement and that the connection between movement and sound was obvious. When asked about the spatialization of sound he had to think for a bit and then said that he felt like he noticed but that it wasn't apparent. The participant felt that the auditory feedback was well synchronized with his movement, especially the tipping sound. He said he did not know if the reason was that the movement was so distinct or that the sound was distinct. He said that the sonification of wheelchair speed (SRS-SM2) also felt distinctly connected to his movements and that the sound produced was *"obvious"*. He added that the other movements did not feel as distinctly synchronized to his movements.

The sonification of turning (SM1) was reported as less obvious by the participant. He said he could notice the sounds when turning but not in the distinct way as for the tipping sound, *"the feeling was that when I tipped [the wheelchair] it was distinct, it felt like playing an instrument. It didn't feel like that when I was turning."* When asked about whether it was the wheelchair or himself producing sound he replied that it was more the wheelchair producing sound than himself. When asked about the consistency in sound for the same movement the participant replied that it sounded the same way when he tipped the wheelchair but that the other sounds were more diffuse. He explained that when he made a turn it did not feel like he could produce a specific sound from a specific kind of turn. When further asked about the sound produced when turning he said that he heard approximately the same sound for the same performed turning. Specifically in the case of exercise 4, the participant reported not hearing a difference in the sound between consecutive similar executions of the exercise, but some difference between different executions. When asked if the auditory feedback could help him find his way back to or reproduce the same movement he replied that it did not, but that the reason probably was that the exercises were too easy for him and that if there was a more complex movement pattern that he had to execute it was more likely that the auditory feedback would help him.

The movement he thought was the hardest to execute was propelling backwards in back wheel balance, in exercise 3, and he said that he would have liked to have more help from the feedback in how straight he was going backwards. The participant also said that during the execution of that exercise he would have been helped more by the tipping sound if it gave information on the acuteness of the tilt, he explicitly requested that the sonification would give information on how close he was to tipping over. When asked if auditory feedback on operation can be meaningful the participant replied that he thought it is meaningful, but that in the context of performing these particular exercises, which were not challenging to him, the sonification system did not give him any additional meaning.

3.3 Interview and Focus Group Summary Novice Users

All participants thought that the sonification was clearly audible. Two of the participants expressed that they occasionally found the feedback to be annoying. All participants understood what movements generated sound with the participant in experiment 3, Male22, pointing out that the initial explanation from the test leader had not been necessary since the mapping was evident after some initial maneuvering.

When asked about the synchronization of movements with sound all participants said they thought it was well synchronized apart from a few short malfunctions, mostly related to the participants moving outside of the area where all rigid body markers are visible to the motion capture cameras. Male28 expressed that the sounds felt well synchronized but that there was information missing, saying, *"there were lots of movements that did not make any sound"*. The group concluded that the difficulty of the exercises made it hard to focus on the sound and thus that some of the experience of coherence in the relation between movement and sound could not be observed. Male28 stated that it was hard to listen to all of the feedback while maneuvering. When asked whether it is the participant or the wheelchair generating sound, Male28 answered it was the wheelchair, Male22 that it was a combination and in the group all participants said they thought it was a cooperation; one participant elaborated and said, *"you put order, the wheelchair is the medium"* and another participant said, *"I make the sound through the wheelchair"*.

Male28 said he thought he would need some more time to practice with the system to tell if the sonification could help him reproduce movements. He continued saying that the sonification of speed, SM2, made it obvious for him how much speed he maintained in exercise 4 and that it helped him in hearing how well he performed between runs. He also expressed that the sonification of tilt, SM3, helped him the most of all the feedback. He said that it helped him to keep an even and good back wheel balance. He also said that the sonification of turning, SM1, did not help him in executing the exercises. Male22 said he thought that it sounded the same every time you performed the same movement, *"when it felt the same, it sounded the same"*, and added that it was the most obvious for the sonification of speed, SM2, in exercise 4. Male28 said he thought it sounded similar every time you performed the same movement. Male22 concluded that the feedback did not help him to reproduce movements in the short time that

the experiment lasted for, but that the sonification of tilt clearly helped him to control the movement.

One participant in the group said she thought that the feedback, in general, does help to control movement. Another participant in the group suggested that trying to repeat a sound they were played beforehand might be easier in reproducing movement because you can focus on the sound only first and then on the sound and movement combination. Male28 made a similar statement.

The spatialization of sound was clearly audible to all of the participants in experiment 4. Male28 did not feel like he could clearly hear that all of the auditory feedback was spatialized, however, after the interview he got to listen when the test leader maneuvered the wheelchair and he then reported that the spatialization was obvious. Male22 said he did not really think about it during the experiment but that it felt like the sound followed him when he turned. He also reported that the spatialization was obvious after listening when the test leader maneuvered the wheelchair.

4. DISCUSSION

If the auditory feedback is experienced as being well synchronized with the wheelchair riders movements and the system as a whole is deemed comprehensible by the user, then there might be reason to believe that the approach is usable in a training context. It is however important to note that this depends on the level of difficulty of the exercises performed. The mapping may be well suited for a specific type of movement but if this movement covers only an intermediate part of the learning process then the system can only be used during that learning window.

The perception of synchronization between movement and sound in the system seems to have been strong. The expert wheelchair rider, Male45, considered all of the feedback to be well synchronized with his movements, however the only part of the sonification that he thought was distinct was the sonification of tilt. It is possible that the reason for that was that the sound of that feedback (SM3) was more distinct. It is also possible that the feeling of getting distinct feedback for that movement was due to the nature of the movement itself. Tilting has a definite start and stop and the feeling of shift of balance is also obvious to Male45, who concluded that the system as is would not help him control movements in the exercises given during the experiment. It seems likely that this is due to the fact that, as Male45 pointed out, the exercises were too easy for him. Some of the exercises that could have been more challenging to him would have been hard to perform in the experiment room because of its relatively small size. The part of the experiment where he had to perform back wheel balance was the only part he considered to be challenging to him. It is interesting that this is also the sonification part that he found to be most useful and where the feedback was most *"distinct"*. So the fact that some novices expressed that they had a feeling of the feedback helping in controlling movement other than tilting could point to that the individual level of difficulty affects the perception of the feedback as help in controlling movement.

In exercise 4, the sonification of turning was tested as well as the sonification of propulsion speed. In that specific context there seemed like the sonification of speed was more successful. While turning sonification could give feedback on the turning characteristics the speed sonification gave a more direct measure on how well the exercise was executed. The participants could clearly hear how much speed was maintained after turning and compare between consecutive executions of the exercise. These two different parameters for sonification then differs not only in its measure of turning performance but also in the time span through which the participant gains insight. The sonification of speed gives a direct measure on how well the exercise was performed. The sonification of turning gives information during the turning operation and it is required of the participant to listen and learn the difference between consecutive runs of the exercise until, after some training, the participant can utilize the feedback for performance improvement. The latter effect could not be observed during these experiments. The fact that the sonification of speed was deemed *"distinct"* by Male45 and useful by both Male28 and Male22 must be seen from the perspective of the above. Though, it is possible that part of the difference is due to the continuity in the sound of the SM1 and SM2, since speed sonification will give continuos feedback while the wheelchair is above the threshold speed while turning sonification will only give feedback during the operation of turning the wheelchair. It could be interesting to remove the threshold for turning sonification. This would mean getting feedback from noise in the system or from only slight turning but would also mean that the participant would get more continuous feedback.

The reason that the tilting sonification was described as more precise and distinct could be that the SM3 sounds more distinct. So it is possible that a more distinct feeling could be acquired from the other parameters mapped if the sound model was changed. The reason for choosing the specific sound models were to produce sounds that were pleasant and that differed from the each other enough to be clearly separated. It is possible that even though the intention was to separate the different parts of the feedback the choice of sound models instead impaired the comprehensibility of the system.

When talking about whether it was the wheelchair or the person generating sound there seemed to be more confidence to say that it was the wheelchair when the person had more skill. All participants concluded, however, that the wheelchair was the sound producing part of the sonification. Since this was apparent, the system can be considered successful in sonifying the wheelchair without sonifying the person. Thus there might be reason to believe that since the sonfication system did not to any large extent help in controlling movements the approach of sonifying the wheelchair while not sonifying the deviation from a good wheelchair movement was not successful for training. However, further research could show an effect on training after multiple sessions with the system.

The system is in general experienced by the participants to give synchronized feedback and in some cases precise and useful feedback that help movement execution. The

approach of sonifying wheelchair movement parameters directly and not in relation to a perfect movement execution requires that the feedback is comprehensible, information rich and well synchronized with the participants movements. If the system is rich enough on information and comprehensible is something that would have to be shown by participants clearly improving their performance after further training with the system. Furthermore, to really validate the system, a number of performance qualities (e.g. movement jitter) and performance metrics (e.g. task completion time) need to be tracked and analysed.

5. CONCLUSION

Sonification of manual wheelchair movements, with direct mapping from movement parameters, does not seem to produce feedback that is meaningful for all types of manual wheelchair operation. It may be used in specific time windows of training and for some movements that are discrete in their nature, as back wheel balance. Participants in this study did however report a good connection between sound and movement when utilizing the system developed which makes it plausible that there is applicability for the system to be used in a training context. It seems likely that for sonification of manual wheelchair operation to be successful it is a good approach to sonify the deviation from a perfect execution rather than direct sonification.

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